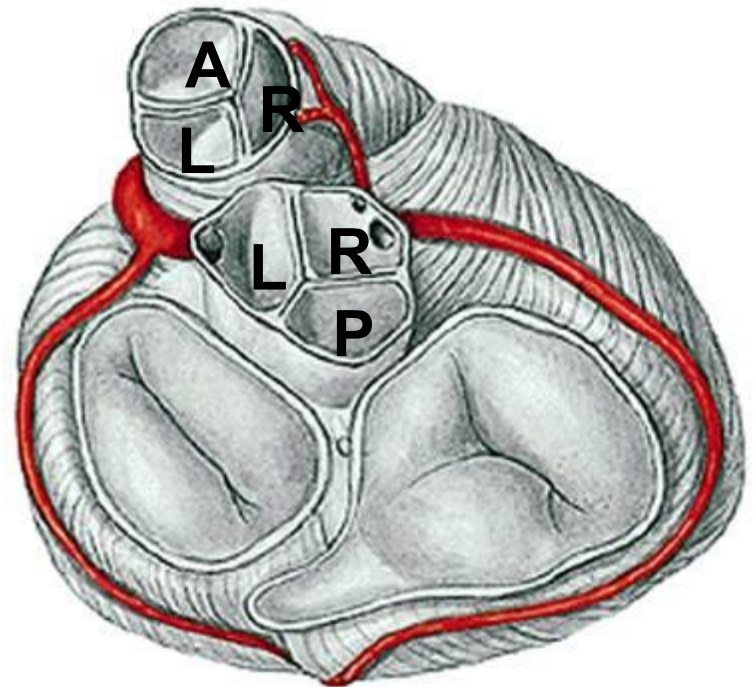
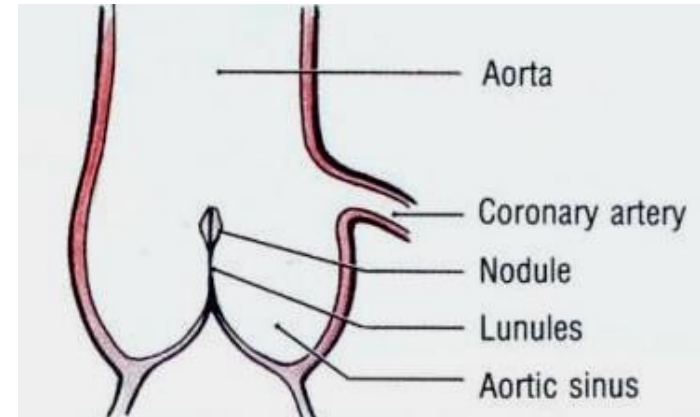
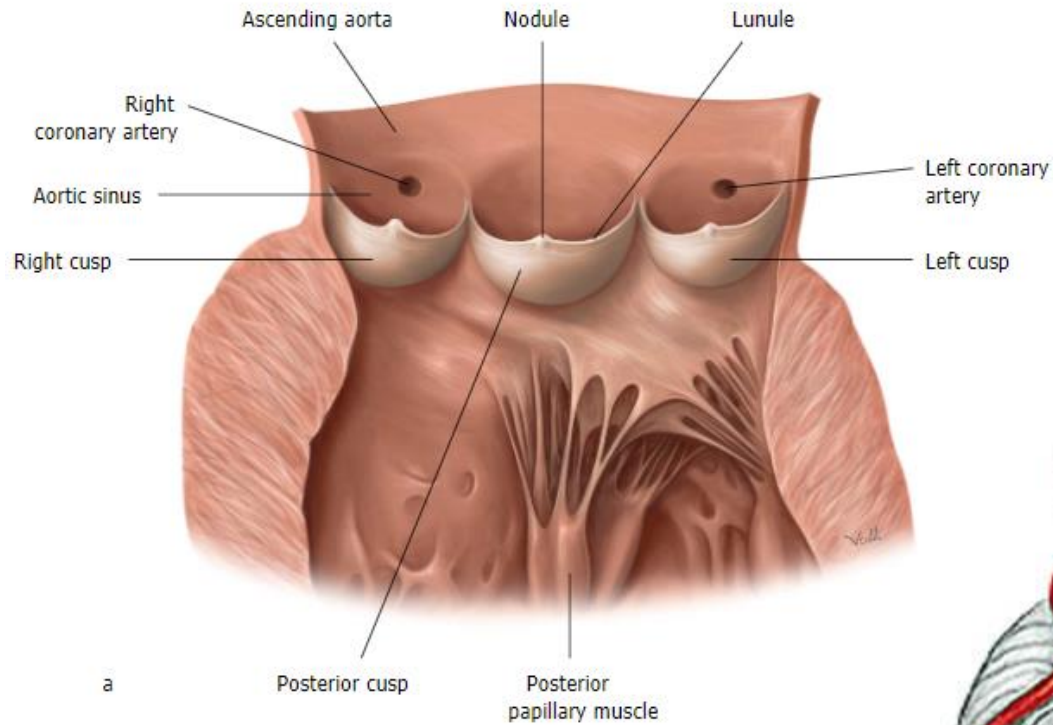


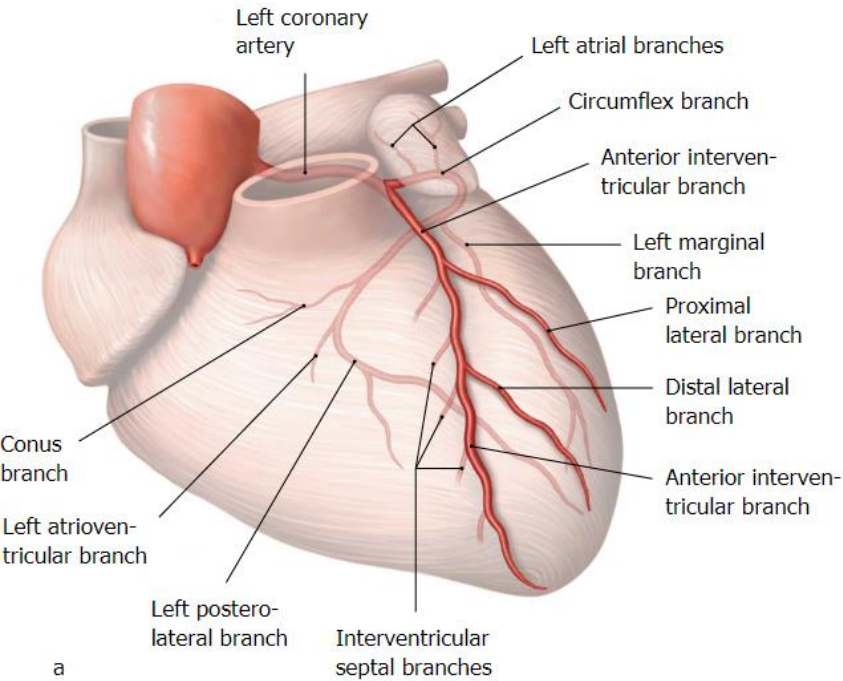
# **Blood supply of the heart I.**

## **Coronary arteries**

# Coronary arteries



# Coronary arteries



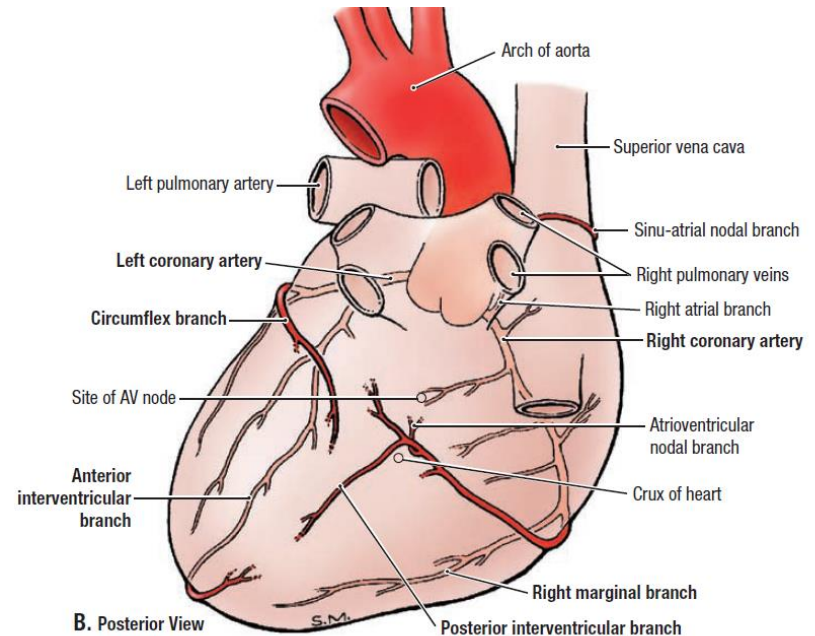
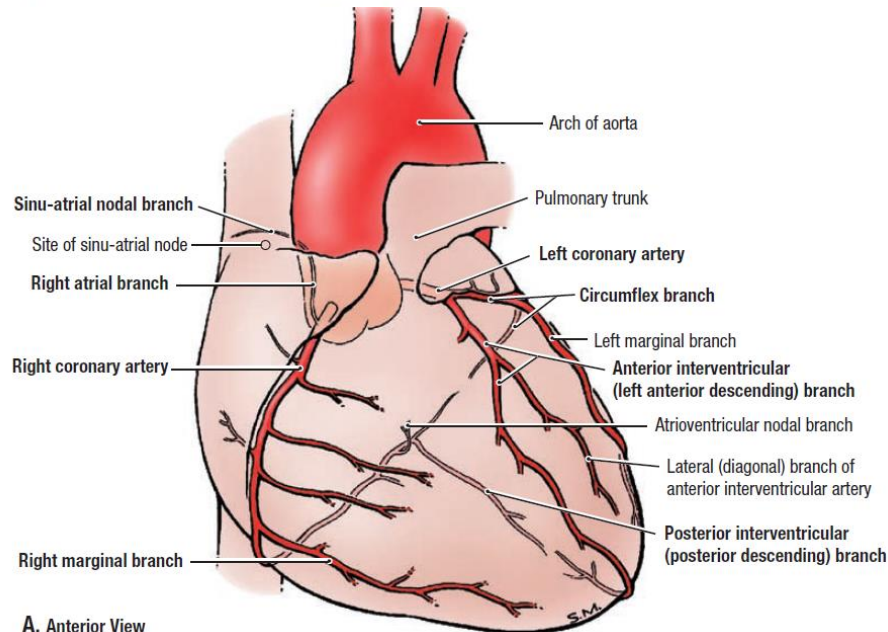
## Left coronary artery (LCA)

### Circumflex branch (LCX)

- Left atrial branches
- Conus branch
- Left marginal branch (LM)
- Left atrioventricular branch (LAV)
- Left posterolateral branch (LPL or PLA) often also the posterior left ventricular branch

### Anterior interventricular branch (AIV or LAD, left anterior descending artery)

- Proximal lateral branch (diagonal branch 1, D1)
- Distal lateral branch (diagonal branch 2, D2)
- Interventricular septal branches

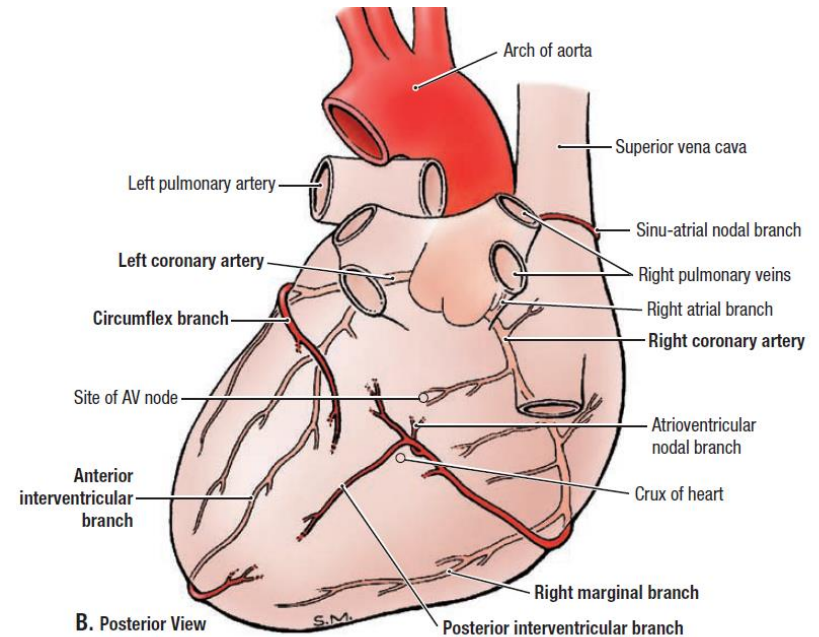
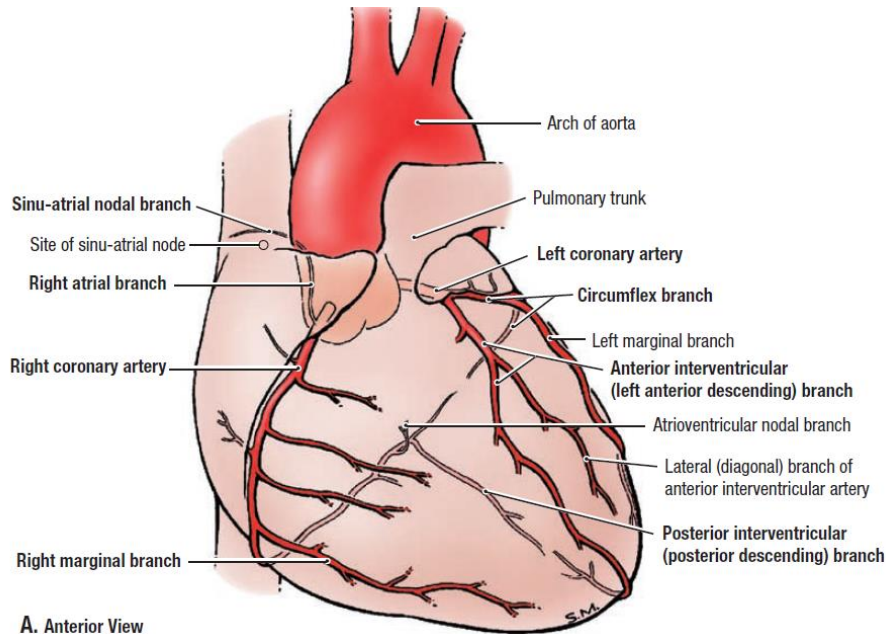
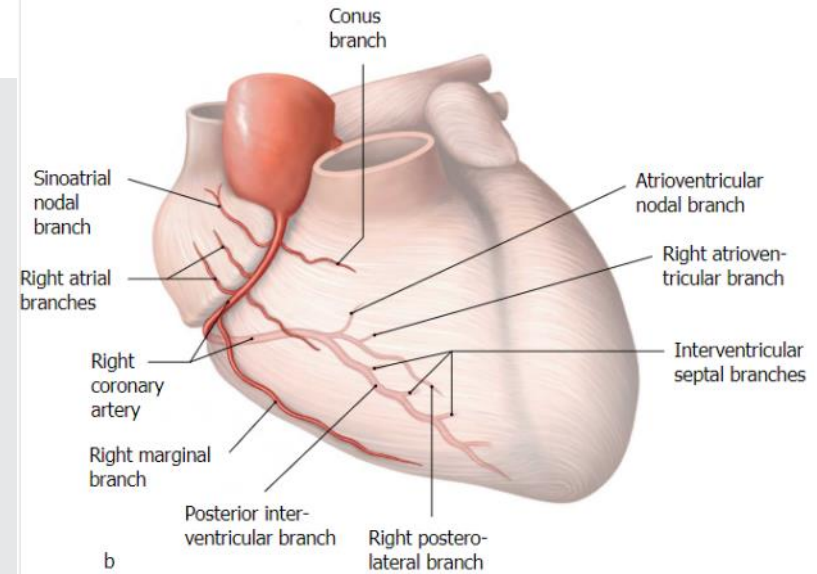




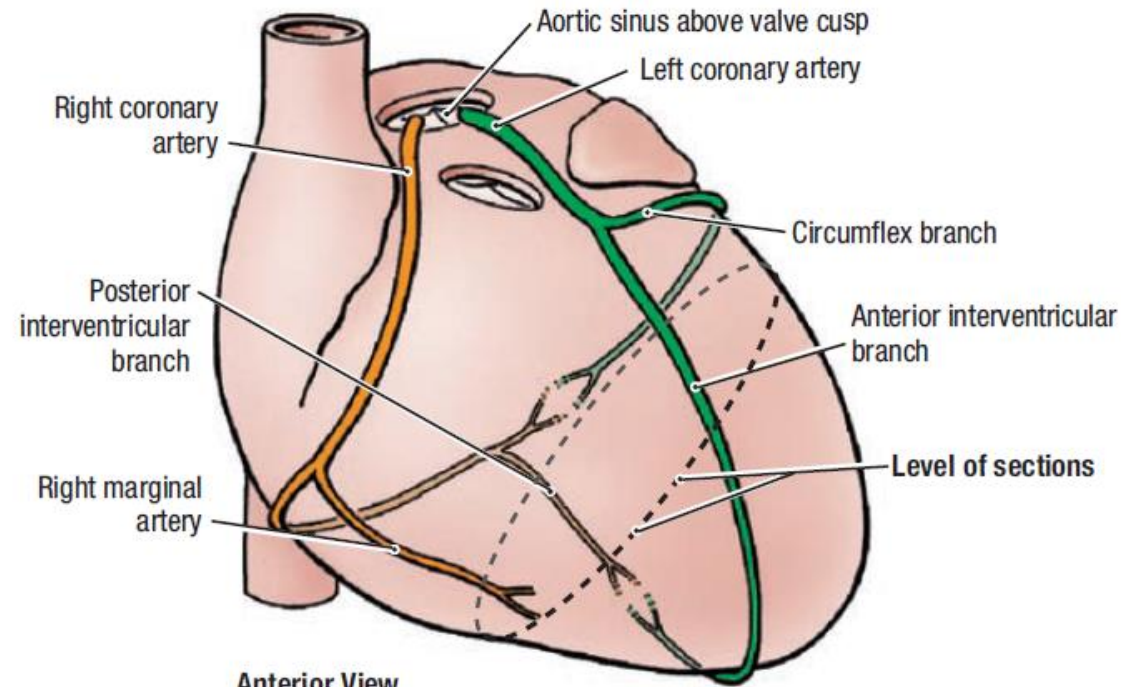
# Coronary arteries

## Right coronary artery (RCA)

- Sinoatrial nodal branch (SAN)
- Right atrial branches
- Conus branch
- Atrioventricular nodal branch (AVN)
- Right marginal branch (RM)
- Posterior interventricular branch (PIV or PDA, posterior descending artery)
- Right atrioventricular branch (RAV)
- Interventricular septal branches
- Right posterolateral branch (RPL or PLA)

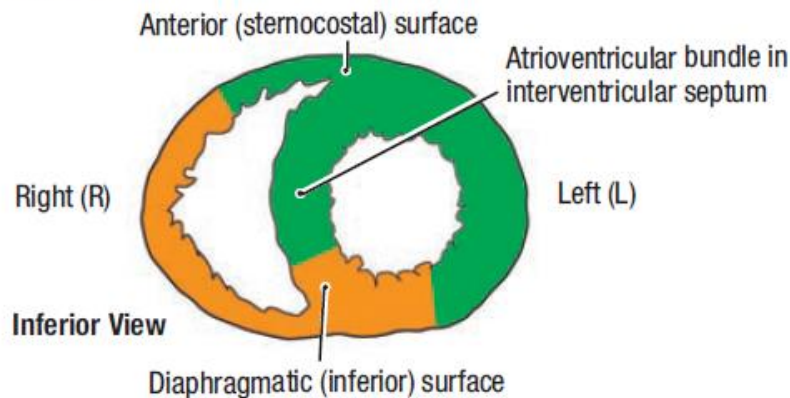


# Coronary arteries, variations



**most common:**

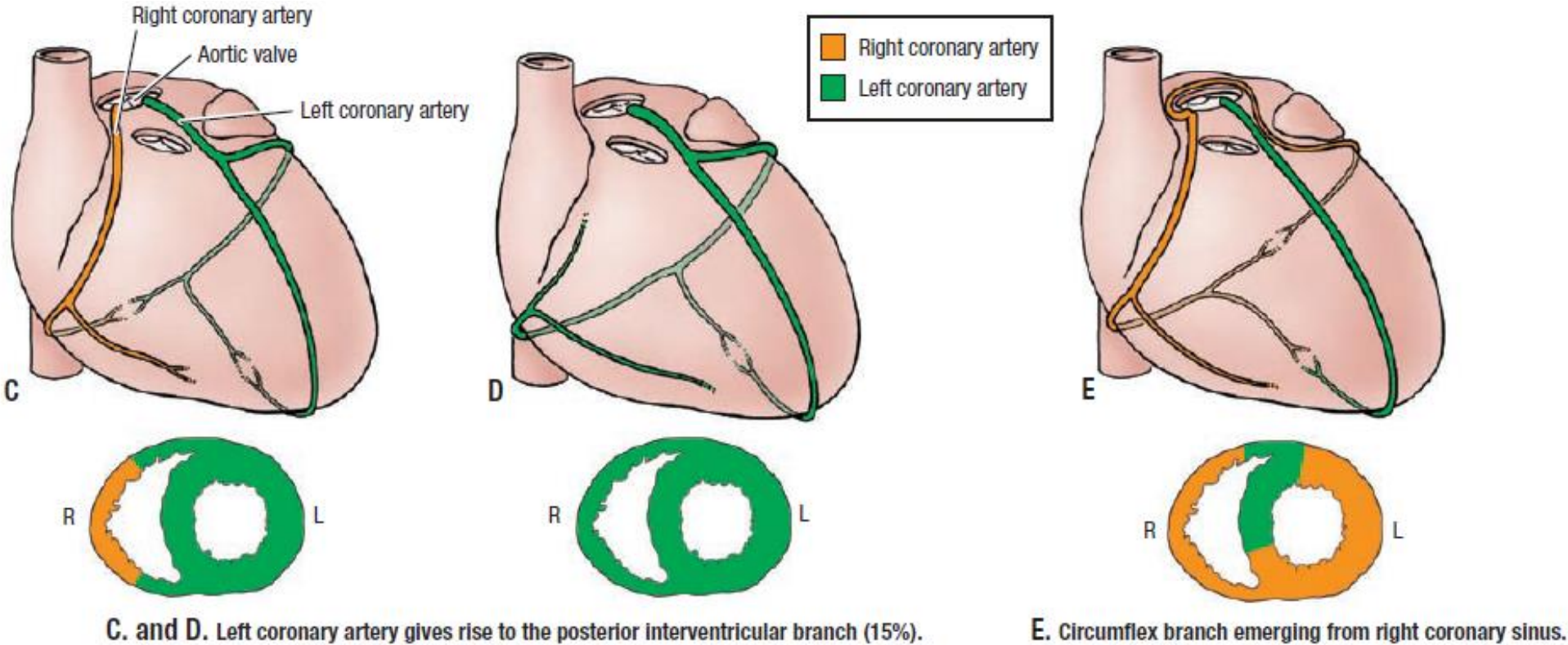
**right coronary dominance: ~50-70%**



**A**

**A. and B.** Most common pattern (67%). Right coronary artery is dominant, giving rise to the posterior interventricular branch.

# Coronary arteries, variations

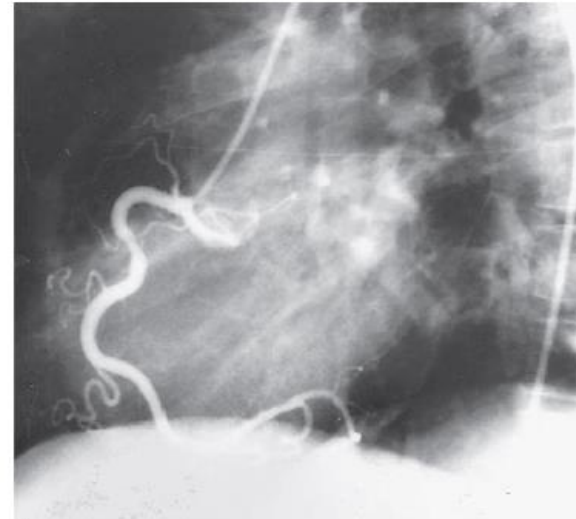
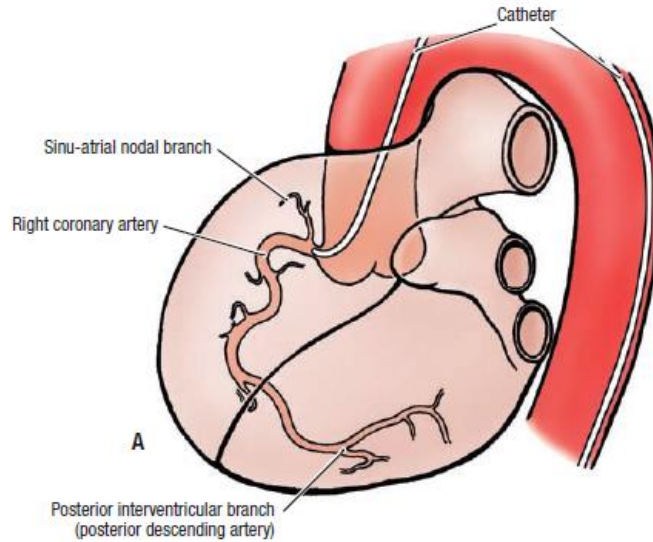


C)-D) left coronary dominance: ~15% (D): only left coronary artery exists )

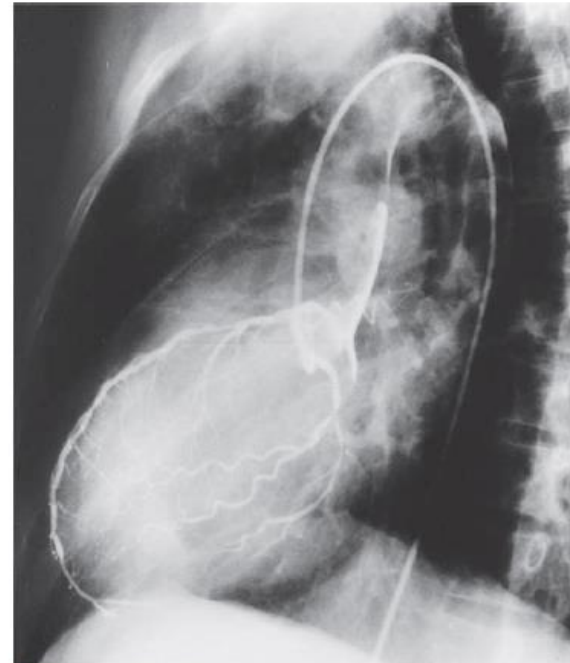
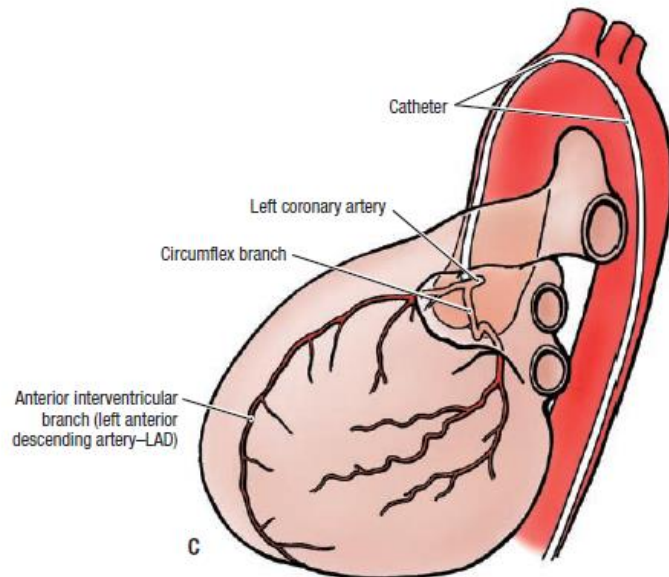
E) circumflex branch arises from right aortic sinus



# Coronary arteries

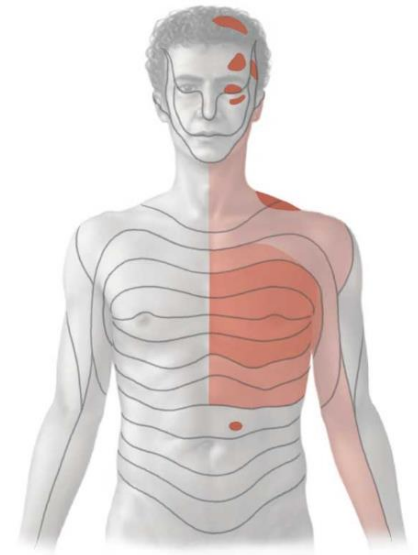
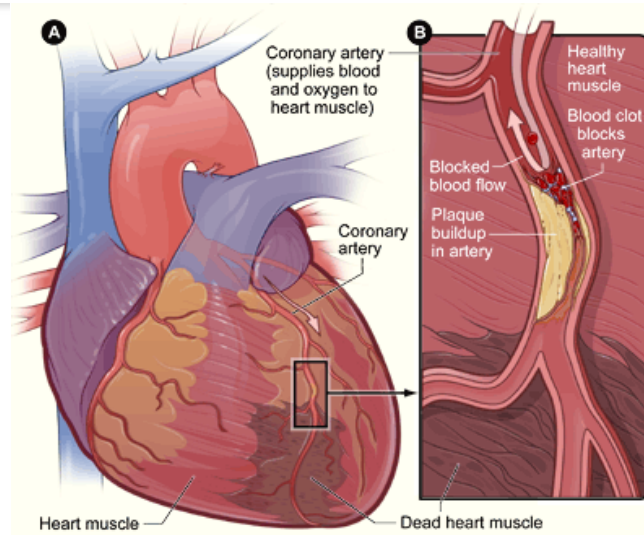
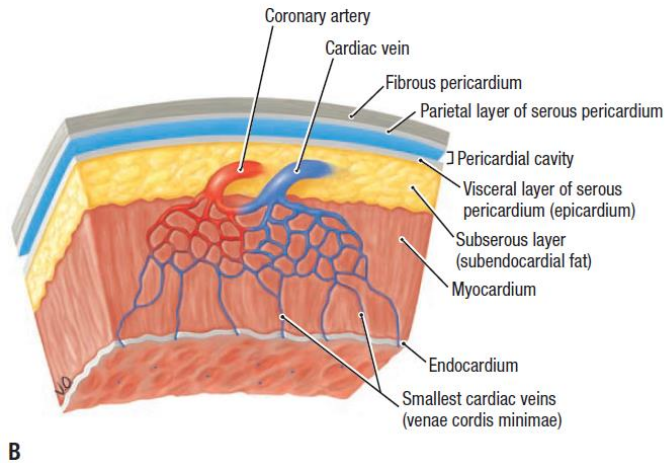


B. Left Anterior Oblique View



D. Left Anterior Oblique View

# Myocardial infarction (heart) attack



myocardial infarction (heart attack): occlusion of the primary or secondary branches of the coronary arteries, the affected myocardium is damaged (dies)

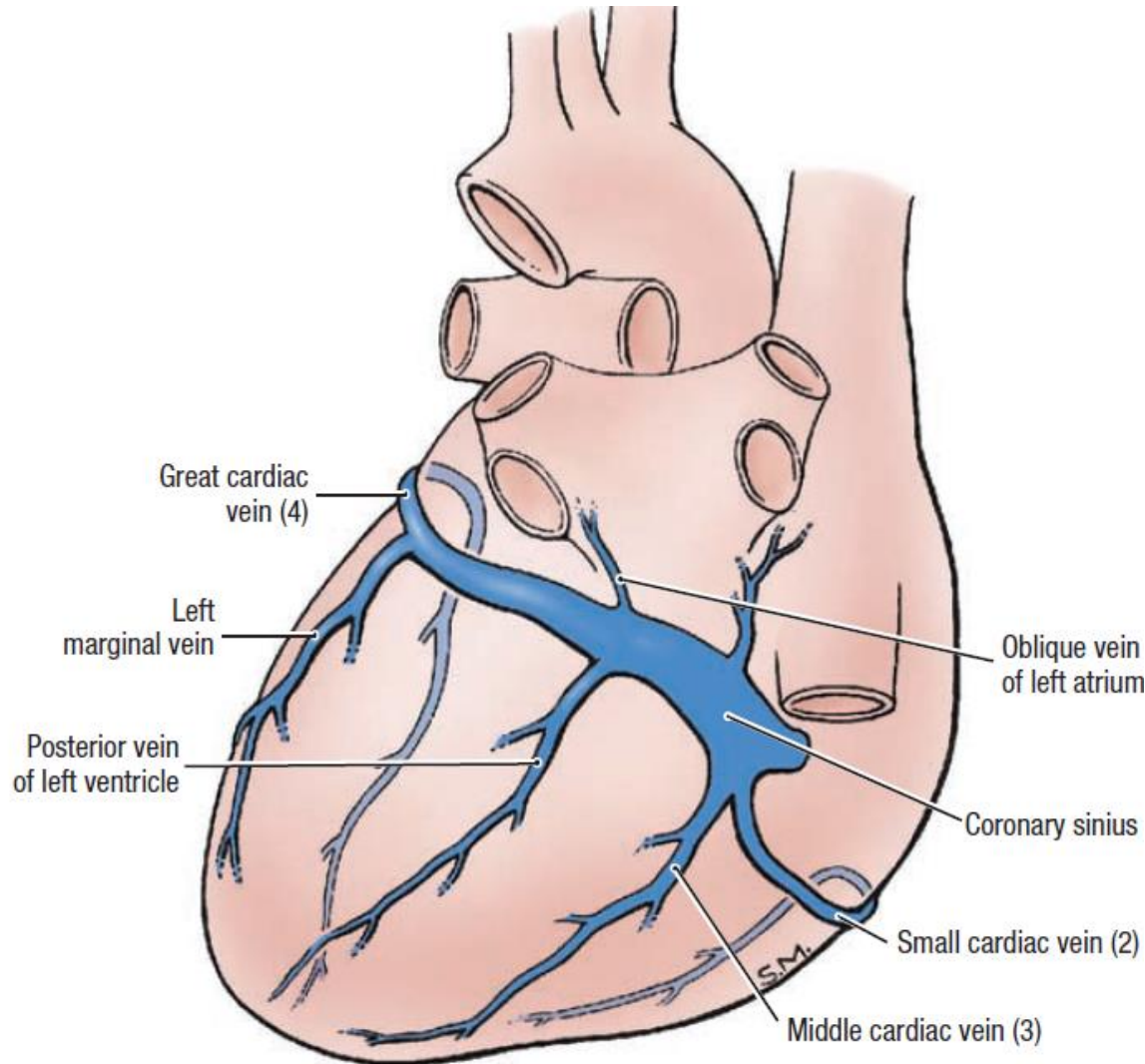
few anastomoses between the coronary artery branches, no collateral circulation



# **Blood supply of the heart II.**

## **Cardiac veins**

# Cardiac veins

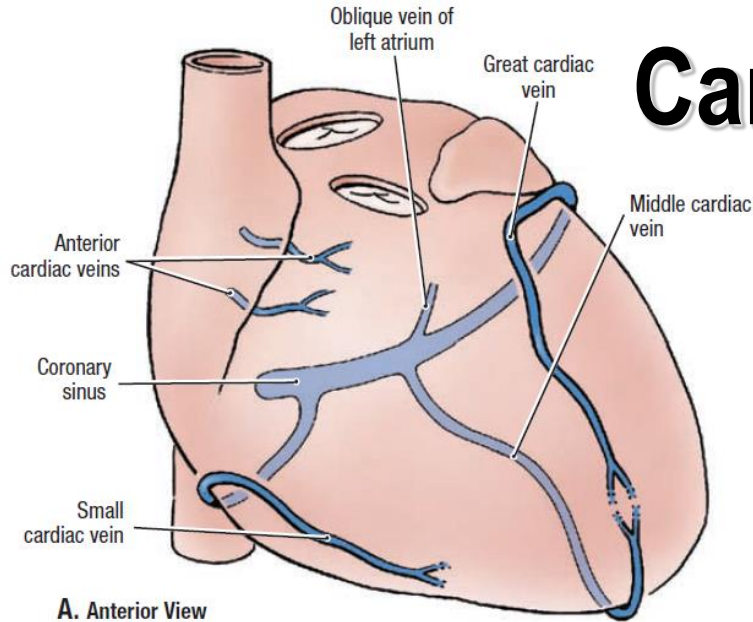


D. Posteroinferior View

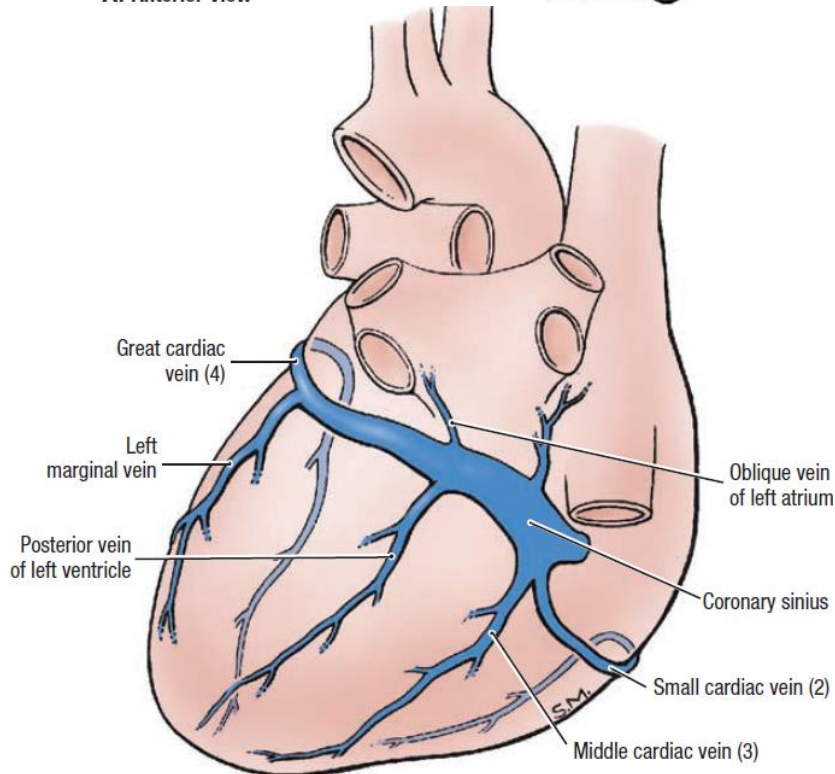
## *Coronary sinus:*

- from left to right in the posterior part of the coronary groove
- opens into the right atrium immediately to the **left of the inf. v. cava, posterior to the right atrioventricular orifice**

# Cardiac veins



A. Anterior View



D. Posteroinferior View

**great cardiac vein:** ant. interventr. groove  
main tributary of the coronary sinus (valve)

**middle cardiac vein:** post. interventr. groove

**small cardiac veins:** coronary groove  
(usually it terminates in the coronary sinus, but it may open directly to the right atrium)

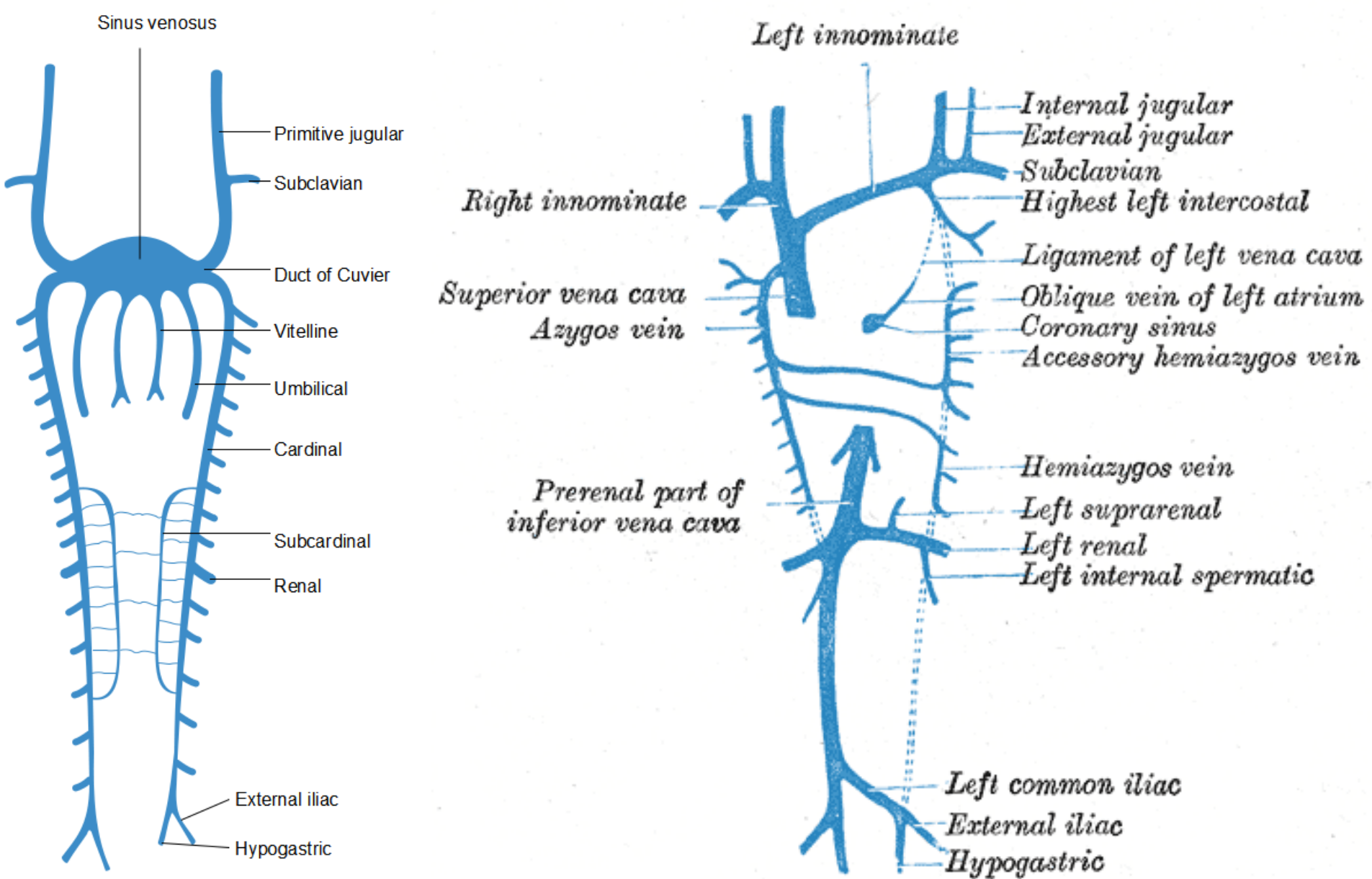
**left posterior cardiac vein:** inf surface  
of the left ventricle

**anterior cardiac veins:** small, ant. surface of the  
right ventricle

**vv. cordis minimae:** directly into the chambers of  
the heart, left atrium

**oblique vein of the left atrium (Marshall vein):**  
post. surface of the left atrium (remnant of left  
ductus Cuvieri)





## Oblique vein of the left atrium (Marshall) development

# Blood supply of the heart - arteries and veins

